NAME: _____ Geometry (FL B.E.S.T.)

8.1 Distance, Midpoint, and Slope

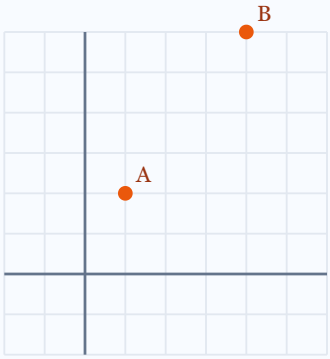
Three formulas describe a segment completely on the coordinate plane. The **distance formula**, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, is just the Pythagorean Theorem applied to horizontal and vertical legs. The **midpoint formula**, $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$, averages the coordinates. The **slope formula**, $m = \frac{y_2 - y_1}{x_2 - x_1}$, measures steepness and direction, and lets you test whether two segments are parallel (equal slopes) or perpendicular (slopes multiply to -1). A horizontal segment has slope 0; a vertical segment has undefined slope. This worksheet also uses the midpoint formula in reverse to find a missing endpoint, and uses slope to test whether three points lie on the same line.

Examples

Example 1: Distance between two points

Find the distance between $A(1, 2)$ and $B(4, 6)$.

Solution



The horizontal and vertical legs are 3 and 4 — a 3-4-5 right triangle.

Identify the coordinates: $(x_1, y_1) = (1, 2)$ and $(x_2, y_2) = (4, 6)$.

Apply the distance formula:

$d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

The points are 5 units apart.

Answer: 5.

Example 2: Midpoint of a segment

Find the midpoint of $(-3, 5)$ and $(7, -1)$.

Solution

Identify the coordinates: $(x_1, y_1) = (-3, 5)$ and $(x_2, y_2) = (7, -1)$.

Average the x-coordinates and the y-coordinates separately:

$$M = \left(\frac{-3 + 7}{2}, \frac{5 + (-1)}{2}\right) = \left(\frac{4}{2}, \frac{4}{2}\right) = (2, 2).$$

This point sits exactly halfway between the two endpoints.

Answer: $(2, 2)$.

Example 3: Slope through two points

Find the slope of the line through $(2, -1)$ and $(5, 8)$.

Solution

Identify the coordinates: $(x_1, y_1) = (2, -1)$ and $(x_2, y_2) = (5, 8)$.

Apply the slope formula:

$$m = \frac{8 - (-1)}{5 - 2} = \frac{9}{3} = 3.$$

A slope of 3 means the line rises 3 units for every 1 unit it runs to the right.

Answer: 3.

Example 4: Find a missing endpoint

$M(3, 1)$ is the midpoint of $\overline{AB}$, with $A(-1, -2)$. Find B.

Solution



M is exactly halfway between A and B.

Solve the midpoint formula for the unknown endpoint: since $x_M = \dfrac{x_A + x_B}{2}$, we get $x_B = 2x_M - x_A$; likewise $y_B = 2y_M - y_A$.

$x_B = 2(3) - (-1) = 6 + 1 = 7.$

$y_B = 2(1) - (-2) = 2 + 2 = 4.$

Answer: $B = (7, 4)$.

Example 5: Testing for parallel lines

Two segments have slopes 2 and 2. Are the lines parallel?

Solution

Two distinct lines with equal slopes never meet — they are parallel.

Answer: Yes, the lines are parallel (assuming they are not the same line).

Example 6: Testing for perpendicular lines

Two segments have slopes 3 and $-\dfrac{1}{3}$. Are the lines perpendicular?

Solution

Multiply the slopes: $3 \times (-\dfrac{1}{3}) = -1.$

A product of -1 confirms perpendicularity.

Answer: Yes.

Example 7: Distance with negative coordinates

Find the distance between $(-3, -2)$ and $(5, 4)$.

Solution

Identify the coordinates: $(x_1, y_1) = (-3, -2)$ and $(x_2, y_2) = (5, 4)$.

Apply the distance formula, being careful with the double negatives:

$d = \sqrt{(5-(-3))^2 + (4-(-2))^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10.$

The two points are 10 units apart.

Answer: 10.

Example 8: Why a horizontal line has slope 0

Explain why every horizontal segment has slope 0, using the points $(-1, 4)$ and $(6, 4)$.

Solution

$$m = \frac{4-4}{6-(-1)} = \frac{0}{7} = 0.$$

The y-coordinates never change along a horizontal line, so the numerator of the slope formula is always 0 — slope 0.

Example 9: Why a vertical line has undefined slope

Explain why every vertical segment has undefined slope, using the points $(3, 7)$ and $(3, -2)$.

Solution

$$m = \frac{-2-7}{3-3} = \frac{-9}{0}.$$

Division by zero is undefined, so a vertical line's slope is undefined — it has no defined "steepness" because it never runs horizontally at all.

Example 10: Are three points collinear?

Determine whether $A(1, 1)$, $B(3, 5)$, and $C(5, 9)$ lie on the same line.

Solution

Slope of $\overline{AB}$: $\dfrac{5-1}{3-1} = \dfrac{4}{2} = 2.$

Slope of $\overline{BC}$: $\dfrac{9-5}{5-3} = \dfrac{4}{2} = 2.$

Both slopes equal 2, and the segments share point B, so all three points lie on the same line.

Answer: Yes, A, B, C are collinear.

Example 11: Application — cell towers on a city grid

Two cell towers sit at $(2, 3)$ and $(14, 11)$ on a city grid measured in kilometers. Find the distance between the towers and the coordinates where a relay station should be built exactly halfway between them.

Solution

Distance: $d = \sqrt{(14-2)^2 + (11-3)^2} = \sqrt{12^2 + 8^2} = \sqrt{144 + 64} = \sqrt{208} = 4\sqrt{13} \approx 14.4$ km.

Midpoint (relay station): $M = \left(\dfrac{2+14}{2}, \dfrac{3+11}{2}\right) = (8, 7).$

Answer: distance $= 4\sqrt{13} \approx 14.4$ km; relay station at $(8, 7)$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Distance

Distance between $(0, 0)$ and $(6, 8)$.

Answer: **Verification:**

Problem 2: Midpoint

Midpoint of $(2, 9)$ and $(-4, 3)$.

Answer: **Verification:**

Problem 3: Slope

Slope through $(-2, 4)$ and $(3, -6)$.

Answer: **Verification:**

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Problem 4: Endpoint

Midpoint $(5, -2)$, $P = (9, 1)$. Find Q.

Answer: **Verification:**

Problem 5: Parallel

Slopes 2 and 2. Parallel?

Answer: **Verification:**

Problem 6: Perpendicular

Slopes $\frac{2}{5}$ and $-\frac{5}{2}$. Perpendicular?

Answer: **Verification:**

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Problem 7: Distance

Between $(1, 1)$ and $(4, 5)$.

Answer: **Verification:**

Problem 8: Midpoint

Of $(0, 0)$ and $(8, -6)$.

Answer: **Verification:**

Problem 9: Slope

Through $(3, 7)$ and $(3, -2)$.

Answer: **Verification:**

Problem 10: Slope

Through $(-1, 4)$ and $(5, 4)$.

Answer: **Verification:**

Problem 11: Endpoint

Midpoint $(0, 0)$, $A = (4, -2)$. Find B.

Answer: **Verification:**

Problem 12: Distance

Between $(-3, 2)$ and $(1, 5)$.

Answer: **Verification:**

Problem 13: Perpendicular

Slopes 1 and 1. Perpendicular?

Answer: **Verification:**

Problem 14: Challenge

$\overline{AB}$ has $A(-3, 4)$ and midpoint $M(1, 1)$. Find endpoint B, then find the length AB.

Answer: **Verification:**

Problem 15: Challenge

Show that $A(0, 0)$, $B(2, 4)$, $C(5, 10)$ are collinear, then verify $AB + BC = AC$.

Answer: **Verification:**

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✓ Answer Key

1. 10
2. $(-1, 6)$
3. -2
4. $(1, -5)$
5. Yes
6. Yes
7. 5
8. $(4, -3)$
9. Undefined
10. 0
11. $(-4, 2)$
12. 5
13. No
14. $B = (5, -2)$; $AB = 10$
15. Collinear (all slopes = 2); $AB = 2\sqrt{5}$, $BC = 3\sqrt{5}$, $AC = 5\sqrt{5}$, and $2\sqrt{5} + 3\sqrt{5} = 5\sqrt{5}$

Great work — keep practising!